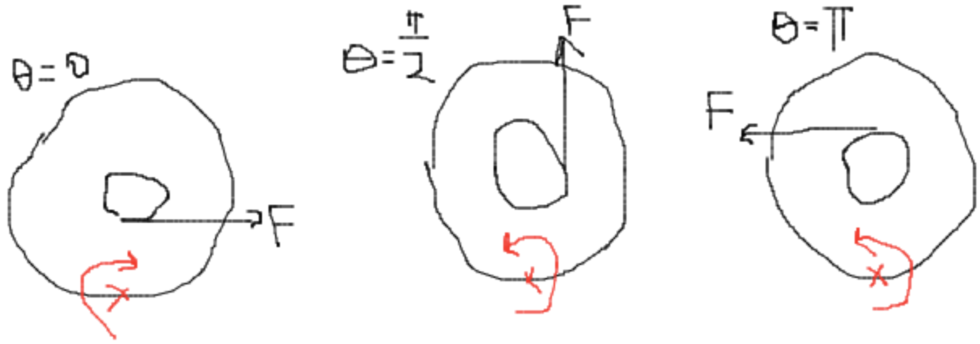
1. (a) With reference to the contact point of the yo-yo with the ground, the net torque is only due to the force F. Drawing the diagrams, we have that the yo-yo rolls clockwise when $\theta = 0$, and anti-clockwise when $\theta = \frac{\pi}{2}$ and $\theta = \pi$.



(b) There should be no net torque about the contact point. The force vector $\vec{F}$ should pass through the contact point. Drawing the diagram, this occurs when $\theta = \cos^{-1}\frac{B}{A}$.

(c) This means friction should be 0. $F_{net} = F\cos\theta = Ma$ and $\tau = F \times B = -\frac{1}{2}MA^2\alpha$. Rolling-without-slipping $a = R\alpha$ gives

$$\cos\theta = -\frac{2BMa}{MA^2\alpha} = -\frac{2B}{A} \therefore \boldsymbol{\theta = \cos^{-1}\left(-\frac{2B}{A}\right)}$$

2. $\dot{x} = v_{COM} = \frac{R}{R+r}v$ and $\theta = 2\tan^{-1}\frac{R}{x}$ thus

$$\dot{\theta} = \frac{2}{1+\left(\frac{R}{x}\right)^2} \cdot \frac{-R}{x^2} \cdot \dot{x} = -\frac{2R^2}{(R^2+x^2)(R+r)}v$$

Since they want the answer in terms of θ, we have:

$$\dot{\theta} = -\frac{2R^2}{\left[R^2 + \left(\frac{R}{\tan\frac{\theta}{2}}\right)^2\right](R+r)}v$$

$$\dot{\theta} = -\frac{2}{\left[1+\cot^2\frac{\theta}{2}\right](R+r)}v$$

$$\dot{\theta} = -\frac{2}{\csc^2\frac{\theta}{2}(R+r)}v = -\frac{\boldsymbol{2\sin^2\frac{\theta}{2}}}{\boldsymbol{(R+r)}}\boldsymbol{v}$$

3. (a) Consider a cylindrical Gaussian surface of radius r and length l.

$$\oiint \vec{E} \cdot \overrightarrow{dA} = 2\pi r l E_r = \frac{\lambda l}{\epsilon_0} \therefore E_r = \frac{\lambda}{2\pi r \epsilon_0}$$

Thus we can integrate to get

$$V = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{b}{a}$$

and capacitance per unit length

$$C_l = \frac{\lambda}{V} = \boldsymbol{\frac{2\pi\epsilon_0}{\ln \frac{b}{a}}}$$

(b) Increasing l by 10% increases C by 10%.

$$C_{new,l\to 1.1l} = 1.1 \frac{2\pi\epsilon_0}{\ln \frac{b}{a}} l$$

But, if a increased by 10%,

$$C_{new,a\to 1.1a} = \frac{2\pi\epsilon_0}{\ln \frac{b}{a} - \ln 1.1} l$$

Comparing:

$$\frac{C_{new,l\to 1.1l}}{C_{new,a\to 1.1a}} = 1.1 \frac{\ln \frac{b}{a} - \ln 1.1}{\ln \frac{b}{a}}$$

which ≥ 1 if and only if

$$\ln \frac{b}{a} \geq \frac{\ln 1.1}{\left(1 - \frac{1}{1.1}\right)} = 11 \ln 1.1$$

so increasing l by 10% is more effective if

$$\boldsymbol{\frac{b}{a} \geq 1.1^{11}}$$

4. (a) Integrating across the spring:

$$\int_0^L \frac{1}{2}\left(\frac{M}{L}dl\right)\left(\frac{vl}{L}\right)^2 = \frac{1}{2}\frac{Mv^2}{L^3}\left(\frac{1}{3}L^3\right) = \boldsymbol{\frac{1}{6}Mv^2}$$

(b) Conservation of energy equation

$$\frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2 = constant$$

which results in angular frequency $\boldsymbol{\omega = \sqrt{k/m}}$.

(c) The new conservation of energy equation

$$\frac{1}{2}m\dot{x}^2 + \frac{1}{6}M\dot{x}^2 + \frac{1}{2}kx^2 = constant$$

which results in angular frequency $\boldsymbol{\omega' = \sqrt{k/(m+M/3)}}$ where $\boldsymbol{M' = m + \frac{M}{3}}$.

5. (a) Dimensional analysis: $[T] = kg\cdot m\cdot s^{-2}, [\rho] = kg\cdot m^{-1}, [v] = m\cdot s^{-1}$

$$[v] = [T]^\alpha[\rho]^\beta = kg^{\alpha+\beta}\cdot m^{\alpha-\beta}\cdot s^{-2\alpha} = m\cdot s^{-1}$$

$$\therefore \begin{cases}\alpha+\beta = 0\\ \alpha-\beta = 1\\ -2\alpha = -1\end{cases} \therefore \boldsymbol{\alpha = \frac{1}{2}, \beta = -\frac{1}{2}} \therefore \boldsymbol{v = T^{1/2}\rho^{-1/2}}$$

(b) The tension at height x above the base of the string is $T = \frac{Mgx}{l} = \rho g x$ so the velocity of the wave is $v = \sqrt{T/\rho} = \sqrt{gx}$. The time taken for the wave to travel down is

$$t = \int_x^l \frac{dx}{v} = \int_x^l \frac{dx}{\sqrt{gx}} = \left[2\sqrt{\frac{x}{g}}\right]_x^l = \frac{2}{\sqrt{g}}\left(\sqrt{l} - \sqrt{x}\right)$$

meanwhile, the time taken for the free-falling body is

$$t = \sqrt{\frac{2(l-x)}{g}}$$

Equating:

$$\frac{2}{\sqrt{g}}\left(\sqrt{l}-\sqrt{x}\right) = \sqrt{\frac{2(l-x)}{g}} \therefore \sqrt{l}-\sqrt{x} = \sqrt{\frac{l-x}{2}} \therefore l + x - 2\sqrt{lx} = \frac{l-x}{2}$$

$$\therefore l + 3x = 4\sqrt{lx} \therefore l^2 + 6lx + 9x^2 = 16lx \therefore l^2 - 10lx + 9x^2 = 0$$

$$\therefore (l-x)(l-9x) = 0 \therefore x = 0 \text{ (rej.) or } x = \boldsymbol{\frac{l}{9}}$$

6. (a) $C = \frac{\epsilon_0 l^2}{d}, V = Ed \therefore Q = CV = \boldsymbol{\epsilon_0 E l^2}$

(b) $E_{due\ to\ one\ plate} = \frac{\sigma}{2\epsilon_0} = \frac{1}{2} E_{due\ to\ both} = \frac{V}{2d}$

$$\therefore F = QE_{due\ to\ one\ plate} = \epsilon_0 \frac{V}{d} l^2 \cdot \frac{V}{2d} = \boldsymbol{\frac{\epsilon_0 V^2 l^2}{2d^2}}$$

(c) $F = mg \therefore \frac{\epsilon_0 V^2 l^2}{2d^2} = \rho l^2 t g$

$$\therefore V = \sqrt{\frac{2\rho d^2 t g}{\epsilon_0}} = \sqrt{\frac{(2 \cdot 2.71 \times 10^3 \cdot (12.7 \times 10^{-3})^2 \cdot (8.00 \times 10^{-6}) \cdot 9.81)}{8.854 \times 10^{-12}}}$$
$$= \mathbf{2780\ V}$$

(d) Energy per unit volume

$$\frac{1}{2}\epsilon_0 E^2 = \frac{1}{2}\epsilon_0 \left(\frac{V}{d}\right)^2 = \frac{\epsilon_0}{2d^2} \cdot \frac{2\rho d^2 t g}{\epsilon_0} = \rho t g = 2.71 \times 10^3 \cdot (8.00 \times 10^{-6}) \cdot 9.81$$
$$= \mathbf{0.213\ J/m^3}$$

7. (a) $dQ = C_P dT \therefore P = C_P \frac{dT}{dt} = C_p T_0 \cdot \frac{1}{3}[1 + a(t - t_0)]^{-2/3} a$

$$\therefore C_p = \frac{3P}{aT_0}[1 + a(t - t_0)]^{2/3} = \frac{3P}{aT_0} \cdot \left(\frac{T}{T_0}\right)^2 = \boldsymbol{\frac{3PT^2}{aT_0^3}}$$

(b) The optical path length is constant.

$$n(r)d + \sqrt{F^2 + r^2} = n_0 d + F \therefore \boldsymbol{n(r) = n_0 - \frac{\sqrt{F^2 + r^2} - F}{d}}$$

Which is $\approx n_0 - \frac{r^2}{2Fd}$, for small r.

8. (a) We know

$$\begin{cases} [x] = m \\ [t] = s \\ [\rho] = kg \cdot m^{-3} \\ [c] = m^2 \cdot s^{-2} \cdot K^{-1} \\ [k] = kg \cdot m \cdot s^{-3} \cdot K^{-1} \end{cases}$$

Some helpful equations to derive the above units are

$$Q = mc\Delta T, Q = kA\frac{\Delta T}{x}$$

We get

$$\begin{cases} \beta + \delta = 0 \\ -3\beta + 2\gamma + \delta = 1 \\ \alpha - 2\gamma - 3\delta = 0 \\ -\gamma - \delta = 0 \end{cases} \therefore \begin{cases} \alpha = 1/2 \\ \beta = -1/2 \\ \gamma = -1/2 \\ \delta = 1/2 \end{cases}$$

(b) We remove factors of c. Initial energy of one K_l meson is $\frac{m(K_l)}{\sqrt{1-\beta^2}} = \sqrt{2}m(K_l)$ and final energy is $\sqrt{m(K_s)^2 + p(K_s)^2}$ thus $p(K_s) = \sqrt{2m(K_l)^2 - m(K_s)^2}$ and $p(K_l) = \frac{m(K_l)\beta}{\sqrt{1-\beta^2}} = m(K_l)$. By setting $\Delta m = m(K_l) - m(K_s) \ll m(K_l)$, we have

$$\Delta p = p(K_s) - p(K_l) = \sqrt{2m(K_l)^2 - m(K_l)^2 + 2m(K_l)\Delta m} - m(K_l) = \Delta m$$

Correcting for the factor of c:

$$\Delta p = 3.5 \times 10^{-6}\text{eV/c}$$

And multiplying by rate of incident mesons:

$$F = rate \cdot \Delta p = 10^6\text{Hz} \cdot 3.5 \times 10^{-6}\text{eV/c} = \mathbf{3.5\ eV/c{\cdot}s}$$

Direction of the force is **opposite direction as the incident beam.**

9. (a) We know the **magnetic field is pointing out of the page** due to right hand rule. By applying Biot-Savart's law:

$$B = \frac{\mu_0}{4\pi}\int \frac{i\left|\overrightarrow{dl} \times \vec{r}\right|}{r^3} = 2\frac{\mu_0 i}{4\pi}\int_0^\infty \frac{\sin\alpha\, d \cdot dl}{(d^2 + l^2 + 2dl\cos\alpha)^{3/2}}$$

$$= 2\frac{\mu_0 i}{4\pi}\int_0^\infty \frac{\sin\alpha\, d \cdot dl}{[d^2\sin^2\alpha + (l + d\cos\alpha)^2]^{3/2}}$$

substitute $l + d\cos\alpha = d\sin\alpha\tan t\,, dl = d\sin\alpha\sec^2 t\, dt$:

$$B = 2\frac{\mu_0 i}{4\pi}\int_A^{\pi/2} \frac{\sin\alpha\, d^2 \sin\alpha \sec^2 t\, dt}{d^3\sin^3\alpha\sec^3 t\, dt} = \frac{\mu_0 i}{2\pi d\sin\alpha}\int_A^{\pi/2} \cos t\, dt = \frac{\mu_0 i(1 - \sin A)}{2\pi d \sin\alpha}$$

where $A = \arctan\frac{\cos\alpha}{\sin\alpha} = \frac{\pi}{2} - \alpha$. Thus

$$B = \frac{\mu_0 i(1 - \cos\alpha)}{2\pi d\sin\alpha} = \frac{\mu_0 i\left(2\sin^2\frac{\alpha}{2}\right)}{2\pi d \cdot 2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}} = \boldsymbol{\frac{\mu_0 i}{2\pi d}\tan\frac{\alpha}{2}}$$

(b) The derivation is similar, replacing $\alpha \coloneqq \pi - \alpha$. The magnetic field is

$$B^* = \boldsymbol{\frac{\mu_0 i}{2\pi d}\cot\frac{\alpha}{2}}$$

pointing into the page.

Comment: Performing the substitution $\alpha \to \pi - \alpha$ works, but $d \to -d$ doesn't!

10. (a) Beta decay,

$${}^{87}_{36}Rb \to {}^{87}_{37}Sr + {}^{0}_{-1}e + \overline{\nu}_e$$

(b) Exponential decay, ${}^{87}Rb(t) = N_0 e^{-\lambda t}, {}^{87}Sr(t) = M_0 + N_0\left(1 - e^{-\lambda t}\right)$ where M_0 is initial amount of ${}^{87}Sr$ and N_0 is initial amount of ${}^{87}Rb$. Let $P_0 = {}^{86}Sr(0) = {}^{86}Sr(t)$. Then $\frac{{}^{87}Sr(0)}{{}^{86}Sr(0)} = \frac{M_0}{P_0}$ is constant while $\frac{{}^{87}Rb(t)}{{}^{86}Sr(t)} = \frac{N_0 e^{-\lambda t}}{P_0}$ varies.

$$\frac{{}^{87}Sr(t)}{{}^{86}Sr(t)} = \frac{M_0 + N_0\left(1 - e^{-\lambda t}\right)}{P_0} = \frac{M_0}{P_0} + \frac{N_0}{P_0}\left(1 - e^{-\lambda t}\right)$$

substitute $\frac{M_0}{P_0} = \frac{{}^{87}Sr(0)}{{}^{86}Sr(0)}, \frac{N_0}{P_0} = \frac{{}^{87}Rb(t)}{{}^{86}Sr(t)}e^{\lambda t}$:

$$\boldsymbol{\frac{{}^{87}Sr(t)}{{}^{86}Sr(t)}} = \frac{{}^{87}Sr(0)}{{}^{86}Sr(0)} + \frac{{}^{87}Rb(t)}{{}^{86}Sr(t)}e^{\lambda t}\left(1 - e^{-\lambda t}\right) = \boldsymbol{\frac{{}^{87}Sr(0)}{{}^{86}Sr(0)} + \frac{{}^{87}Rb(t)}{{}^{86}Sr(0)}\left(e^{\lambda t} - 1\right)}$$

(c) $e^{\lambda\tau_m} - 1 = \frac{0.0125}{0.25} = 0.05 \therefore \tau_m = \frac{\ln 1.05}{\lambda} = \frac{\ln 1.05}{\ln 2}T_{1/2} = \boldsymbol{3.4 \times 10^9}$ **years**